Doojav (2018) Model Modified for DSTI: Equation-by-Equation Notes

Generated explanation for the DSTI-modified Dynare file

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How to read the model

The Dynare file uses `model(linear)`, so each endogenous variable is interpreted as a linear deviation from steady state, or as the linearized counterpart of the original nonlinear variable. Time subscripts are written as t, with x_{t-1} for `x(-1)` and x_{t+1} for `x(+1)`.

The main DSTI modifications are:

- household credit enters consumption directly through $prm\_crhh \cdot cr_{hh,t}$;
- household credit responds to the DSTI instrument through $\lambda_{5,hh} dsti_t$;
- a DSTI policy rule is added as Equation (40-1).

Households

Equation (1): Forward-looking consumption IS equation

$$c_t - hc_{t-1} = (c_{t+1} - hc_t) - \frac{1-h}{\sigma}\left[\alpha_p(r_t - \pi_{t+1}) + (1-\alpha_p)(rl_t - \pi_{t+1})\right]$$
$$+ \frac{\sigma-1}{\sigma}\bar{wlc}(n_t - n_{t+1}) + prm\_crhh\, cr_{hh,t} + \varepsilon_{c,t}.$$
$$\varepsilon_{c,t} = u_{g,t} + u_{c,t}.$$

This is the household Euler/IS relation with habit formation. Current consumption relative to habit depends positively on expected future consumption and negatively on expected real borrowing costs. The policy rate r_t matters for patient households, while the lending rate rl_t matters for credit-constrained households. The new term $prm\_crhh\, cr_{hh,t}$ makes household credit directly raise consumption demand. The consumption disturbance is now driven by fiscal and preference shocks, not by the household-credit shock directly.

Equation (2): Uncovered interest parity with risk premium

$$r_t - r_t^\star = (1-\phi_{ex})d\_exc_{t+1} - \phi_{ex} d\_exc_t - \phi_{asst} asst_t + \varepsilon_{uip,t}.$$
$$\varepsilon_{uip,t} = \rho_{uip}\varepsilon_{uip,t-1} + u_{uip,t}.$$

The interest-rate differential is linked to expected and current nominal depreciation. A higher net foreign asset position lowers the risk premium term when $\phi_{asst} > 0$. The AR(1) process captures persistent UIP or country-risk shocks.

Equation (3): Nominal exchange-rate depreciation identity

$$d\_exc_t = \pi_t - \pi_t^\star + d\_q_t.$$

Nominal depreciation equals the domestic-foreign inflation differential plus the change in the real exchange rate. This is a log-linear purchasing-power-parity accounting identity.

Equation (4): Real exchange rate

$$q_t = \psi_{f,t} + (1 - \alpha_{fc}) s_t.$$

The real exchange rate combines the foreign law-of-one-price gap ψ_f and the terms of trade s_t. The weight $1 - \alpha_{fc}$ reflects the domestic component in the consumption basket.

Equation (5): Law-of-one-price gap

$$\psi_{f,t} = exc_t + p_t^\star + p_{f,t}.$$

This defines the gap between the domestic-currency price of foreign goods and the corresponding domestic import-price index, using the sign convention in the code. It links the nominal exchange-rate level, foreign price level, and import price level.

Equation (6): Effective capital supply

$$k_{s,t} = k_{t-1} + utlz_t.$$

Capital services supplied to firms equal predetermined installed capital plus the utilization margin. Higher utilization raises effective capital even if the physical capital stock is fixed in the short run.

Equation (7): Capital accumulation

$$k_t = (1 - \delta) k_{t-1} + \delta inv_t.$$

The capital stock evolves with depreciation and investment. Since the model is linearized, δ controls the speed with which investment affects the capital stock.

Equation (8): Investment demand

$$inv_t = \alpha_{fdi}(fdi_t + q_t) + (1 - \alpha_{fdi}) cr_{frm,t} + \varepsilon_{inv,t}.$$

$$\varepsilon_{inv,t} = \rho_{inv} \varepsilon_{inv,t-1} + u_{inv,t}.$$

Investment is financed by a weighted combination of foreign direct investment and domestic firm credit. A real exchange-rate movement affects the domestic value of FDI. The investment shock captures persistent investment-specific disturbances.

Equation (9): Foreign direct investment process

$$fdi_t = \rho_{fdi} fdi_{t-1} + u_{fdi,t}.$$

FDI is modeled as an exogenous persistent AR(1) process. In the active shocks block, u_{fdi} is the only shock assigned a nonzero standard error.

Equation (10): Capital utilization

$$utlz_t = \alpha_{utlz} r_{k,t}.$$

Firms utilize capital more intensively when the rental rate of capital rises. The parameter $\alpha_{utlz} = (1-\kappa)/\kappa$ controls the elasticity of utilization with respect to the rental return.

Firms and Prices

Equation (11): Domestic production

$$y_t = \alpha_{ks} k_{s,t} + (1-\alpha_{ks}) n_t + \varepsilon_{tfp,t}.$$

$$\varepsilon_{tfp,t} = \rho_{tfp} \varepsilon_{tfp,t-1} + u_{tfp,t}.$$

Output is produced with capital services and labor. α_{ks} is the capital share. The TFP shock shifts production independently of factor inputs.

Equation (12): Capital rental rate

$$r_{k,t} = -(k_{s,t} - n_t) + w_t = w_t + n_t - k_{s,t}.$$

The rental rate of capital rises when wages or labor input rise relative to capital services. Intuitively, scarce capital relative to labor commands a higher rental return.

Equation (13): Real marginal cost

$$mc_t = (1-\alpha_{ks}) w_t + \alpha_{ks} r_{k,t} \left[(1-\alpha_{ks})\upsilon_n + \alpha_{ks}\upsilon_k\right] r_t \alpha_{fc} s_t - \varepsilon_{tfp,t}.$$

Marginal cost is a weighted average of labor cost, capital cost, financing cost, and terms-of-trade pressure. Higher productivity reduces marginal cost because firms can produce more from the same inputs.

Equation (14): Domestic inflation Phillips curve

$$\pi_{h,t} - \gamma_h \pi_{h,t-1} = \beta(\pi_{h,t+1} - \gamma_h \pi_{h,t}) + \kappa_h mc_t + \varepsilon_{pih,t}.$$

$$\varepsilon_{pih,t} = \rho_{pih} \varepsilon_{pih,t-1} + u_{pih,t}.$$

Domestic inflation depends on past indexation, expected future inflation, and current real marginal cost. The slope $\kappa_h = ((1-\theta_h)(1-\theta_h\beta))/\theta_h$ is higher when domestic prices are more flexible.

Equation (15): Import inflation Phillips curve

$$\pi_{f,t} - \gamma_f \pi_{f,t-1} = \beta(\pi_{f,t+1} - \gamma_f \pi_{f,t}) + \kappa_f(\psi_{f,t} + \upsilon_f r_t) + \varepsilon_{pif,t}.$$

$$\varepsilon_{pif,t} = \rho_{pif} \varepsilon_{pif,t-1} + u_{pif,t}.$$

Import inflation follows a New Keynesian Phillips curve for imported goods. The law-of-one-price gap and monetary/financing component affect import-price pressure.

Equation (16): Domestic input for consumption

$$c_{h,t} = c_t + \eta \alpha_{fc} s_t.$$

Demand for domestic consumption goods rises with aggregate consumption and with the terms of trade according to the substitution elasticity η. The import share α_{fc} governs the exposure.

Equation (17): Imported input for consumption

$$c_{f,t} = c_t + \eta(1 - \alpha_{fc})s_t.$$

Imported consumption demand also rises with aggregate consumption and reacts to relative-price movements. The weight is complementary to the domestic input weight.

Equation (18): Consumer price inflation

$$\pi_t = \pi_{h,t} + \alpha_{fc} ds_t.$$

CPI inflation equals domestic inflation plus the import-share-weighted change in the terms of trade. When imported goods become relatively more expensive, CPI inflation rises.

Equation (19): Change in terms of trade from inflation gap

$$ds_t = \pi_{f,t} - \pi_{h,t}.$$

The change in the terms of trade equals import inflation minus domestic inflation.

Equation (20): Investment input

$$i_{h,t} = inv_t.$$

The domestic intermediate input used for investment is set equal to aggregate investment. This is an accounting simplification.

Equation (21): Non-resource exports

$$xpt\_nr_t = \eta(s_t + \psi_{f,t}) + y_t^{\star}.$$

Non-resource exports increase with foreign demand $y_t^{\star}$ and relative-price competitiveness. The elasticity η controls the price response.

Equation (22): Resource exports

$$xpt\_r_t = y_t^{\star} + \varepsilon_{xptr,t}.$$

$$\varepsilon_{xptr,t} = \rho_{xptr}\varepsilon_{xptr,t-1} + u_{xptr,t}.$$

Resource exports move with foreign activity and a persistent resource-export shock.

Labor Market

Equation (23): Wage markup

$$\mu_{w,t} = w_t - (z_t + \varphi n_t + \varepsilon_{n,t}).$$

$$\varepsilon_{n,t} = \rho_n \varepsilon_{n,t-1} + u_{n,t}.$$

The wage markup is the gap between the real wage and the household's reference wage schedule. Higher labor supply n_t, the reference shifter z_t, or the labor-supply shock lowers the markup for a given wage.

Equation (24): Unemployment relation

$$\mu_{w,t} = \varphi un_t.$$

Unemployment is proportional to the wage markup. A higher markup implies that wages are high relative to labor-market fundamentals, generating unemployment.

Equation (25): Endogenous reference shifter

$$z_t = (1 - \vartheta_z) z_{t-1} + \vartheta_z \left[-\varepsilon_{c,t} + \frac{\sigma}{1-h}(c_t - hc_{t-1}) \right].$$

The reference shifter updates gradually. It rises when consumption relative to habit is high, and falls when the consumption shock is positive, according to the sign convention in the model.

Equation (26): Real wage law of motion

$$w_t = w_{t-1} + \pi_{w,t} - \pi_t.$$

The real wage rises when nominal wage inflation exceeds consumer price inflation.

Equation (27): Wage Phillips curve

$$\pi_{w,t} - \gamma_w \pi_{w,t-1} = \beta(\pi_{w,t+1} - \gamma_w \pi_{w,t}) + \kappa_w(\mu_{w,t} - \mu_{w,n,t}).$$

$$\varepsilon_{piw,t} = \rho_{piw} \varepsilon_{piw,t-1} + u_{piw,t}.$$

Wage inflation depends on indexation, expected future wage inflation, and the gap between the wage markup and its shock-driven benchmark $\mu_{w,n,t}$. The wage shock enters through Equation (49).

Equation (28): Labor force identity

$$l_t = n_t + un_t.$$

The labor force equals employment plus unemployment.

Financial Market

Equation (29): Aggregate credit

$$cr_t = \zeta cr_{hh,t} + (1 - \zeta) cr_{frm,t}.$$

Total credit is a weighted average of household credit and firm credit. The parameter ζ is the household-credit share.

Equation (30): Household credit demand and policy constraint

$$cr_{hh,t} = \lambda_{1,hh} y_t - \lambda_{2,hh} rl_t - \lambda_{3,hh} c\_rtio_t - \lambda_{4,hh} rr_t + \lambda_{5,hh} dsti_t + \varepsilon_{crhh,t}.$$

$$\varepsilon_{crhh,t} = \rho_{crhh} \varepsilon_{crhh,t-1} + u_{crhh,t}.$$

Household credit rises with output and falls when lending rates, capital requirements, or reserve requirements rise. The DSTI term is new. With $\lambda_{5,hh} > 0$, a higher $dsti_t$ relaxes household borrowing conditions and raises household credit. If $dsti_t$ is interpreted as policy tightness rather than an allowed DSTI ceiling, the sign would need to be reconsidered.

Equation (31): Firm credit

$$cr_{frm,t} = \lambda_{1,frm} y_t - \lambda_{2,frm} rl_t - \lambda_{3,frm} c\_rtio_t - \lambda_{4,frm} rr_t + \varepsilon_{crfrm,t}.$$

$$\varepsilon_{crfrm,t} = \rho_{crfrm} \varepsilon_{crfrm,t-1} + u_{crfrm,t}.$$

Firm credit increases with output and decreases with borrowing rates and macroprudential requirements. The AR(1) shock captures persistent firm-credit disturbances.

Equation (32): Bank lending rate

$$rl_t - rl_{t-1} = \beta(rl_{t+1} - rl_t) - \kappa_{rl}(rl_t - cst\_fnd_t) + \varepsilon_{rl,t}.$$

$$\varepsilon_{rl,t} = \rho_{rl} \varepsilon_{rl,t-1} + u_{rl,t}.$$

The lending rate adjusts gradually toward banks' funding cost. When rl_t is above funding cost, the negative adjustment term pushes it down.

Equation (33): Cost of funding

$$cst\_fnd_t = r_t + \nu_1 npl_t + \nu_2 c\_rtio_t + \nu_3 rr_t.$$

Bank funding cost rises with the policy rate, nonperforming loans, capital requirements, and reserve requirements.

Equation (34): Aggregate nonperforming loans

$$npl_t = \gamma_1 npl_{hh,t} + \gamma_2 npl_{frm,t}.$$

Total NPLs are a weighted average of household and firm NPLs.

Equation (35): Household nonperforming loans

$$npl_{hh,t} = \xi_{hh1} npl_{hh,t-1} - \xi_{hh2} y_t + \xi_{hh3} d\_exc_t + \varepsilon_{nplhh,t}.$$

$$\varepsilon_{nplhh,t} = \rho_{nplhh} \varepsilon_{nplhh,t-1} + u_{nplhh,t}.$$

Household NPLs are persistent, fall when output improves, and rise with nominal depreciation. Depreciation may worsen repayment capacity through foreign-currency balance-sheet effects.

Equation (36): Firm nonperforming loans

$$npl_{frm,t} = \xi_{frm1} npl_{frm,t-1} - \xi_{frm2} y_t + \xi_{frm3} d\_exc_t + \varepsilon_{nplfrm,t}.$$

$$\varepsilon_{nplfrm,t} = \rho_{nplfrm} \varepsilon_{nplfrm,t-1} + u_{nplfrm,t}.$$

Firm NPLs follow the same logic as household NPLs: persistence, lower defaults in expansions, and higher defaults after depreciation.

Fiscal, Monetary, and Macroprudential Policy

Equation (37): Government spending

$$g_t = \iota_g g_{t-1} + \iota_{gfdi} fdi_t + \iota_{gx} xpt\_r_t + \iota_{gcop} prc\_cmm_t + u_{g,t}.$$

Government spending is persistent and responds to FDI, resource exports, and commodity prices. This captures fiscal dependence on external and commodity conditions.

Equation (38): Monetary policy rule

$$r_t = \rho_r r_{t-1} + (1 - \rho_r)(\chi_\pi \pi_t + \chi_y y_t + \chi_{dexc} d\_exc_t) + u_{r,t}.$$

The central bank smooths the policy rate and responds to inflation, output, and exchange-rate depreciation.

Equation (39): Capital requirement rule

$$c\_rtio_t = \bar{\omega}_1 c\_rtio_{t-1} + (1 - \bar{\omega}_1)(\bar{\omega}_2 y_t + \bar{\omega}_3 cr_t) + u_{crtio,t}.$$

The capital requirement can be made countercyclical by responding to output and credit. In the provided calibration, $\bar{\omega}_1 = \bar{\omega}_2 = \bar{\omega}_3 = 0$, so the rule is inactive except for the shock u_{crtio}.

Equation (40): Reserve requirement rule

$$rr_t = \upsilon_1 rr_{t-1} + (1 - \upsilon_1)(\upsilon_2 y_t + \upsilon_3 cr_t) + u_{rr,t}.$$

Reserve requirements can respond to output and credit. In the provided calibration, $\upsilon_1 = \upsilon_2 = \upsilon_3 = 0$, so the endogenous rule is also inactive except for the shock u_{rr}.

Equation (40-1): DSTI policy rule

$$dsti_t = \varepsilon_1 dsti_{t-1} + (1 - \varepsilon_1)(\varepsilon_2 \pi_{w,t} - \varepsilon_3 cr_{hh,t}) + u_{dsti,t}.$$

The DSTI instrument is persistent. It rises with wage inflation and falls when household credit is high. Combined with Equation (30), where a higher DSTI ceiling raises household credit, this creates a stabilizing feedback: strong household credit lowers the future DSTI setting, which then restrains household borrowing.

Foreign Variables and Commodity Price

Equation (41): Foreign output

$$y_t^\star = \rho_{ystar} y_{t-1}^\star + u_{ystar,t}.$$

Foreign output is an exogenous AR(1) process.

Equation (42): Foreign inflation

$$\pi_t^\star = \rho_{pistar} \pi_{t-1}^\star + u_{pistar,t}.$$

Foreign inflation is exogenous and persistent.

Equation (43): Foreign interest rate

$$r_t^\star = \rho_{rstar} r_{t-1}^\star + u_{rstar,t}.$$

The foreign policy rate is exogenous and persistent.

Equation (44): Foreign commodity price

$$prc\_cmm_t^\star = \rho_{pcmmstar} prc\_cmm_{t-1}^\star + u_{pcmmstar,t}.$$

The foreign commodity price follows an AR(1) process.

Resource Constraint and External Accounts

Equation (45): Goods-market equilibrium

$$y_t = c_y c_t + g_y g_t + i_y i_{h,t} + xptnr_y xpt\_nr_t + xptr_y xpt\_r_t + utlz_y utlz_t.$$

Output is allocated across consumption, government spending, investment, non-resource exports, resource exports, and utilization costs. The coefficients are steady-state shares.

Equation (46): Net foreign assets

$$\begin{aligned} asst_t =& \frac{1}{\beta} asst_{t-1} + xptnr_y(xpt\_nr_t - \alpha_{fc} s_t) \\ &+ xptr_y(q_t + prc\_cmm_t + xpt\_r_t) - m_y(q_t + c_{f,t}) + fdi_y(q_t + fdi_t). \end{aligned}$$

Net foreign assets evolve with interest accumulation and the external balance. Non-resource exports, resource exports, imports, and FDI all affect the external position, with real exchange-rate and price adjustments included.

Equation (47): Domestic commodity price

$$prc\_cmm_t = \alpha_r prc\_cmm_t^\star + (1 - \alpha_r) prc\_cmm_{t-1}.$$

The domestic commodity price is a weighted average of the foreign commodity price and its own lag. α_r controls pass-through from world commodity prices.

Extra Linking Equations

Equation (48): Terms-of-trade level and change

$$ds_t = s_t - s_{t-1}.$$

This links the change in the terms of trade to the level s_t. Together with Equation (19), it implies that import inflation relative to domestic inflation drives the terms-of-trade level.

Equation (49): Wage markup shock benchmark

$$\mu_{w,n,t} = 100\, \varepsilon_{piw,t}.$$

The wage shock is scaled and introduced into the wage Phillips curve through the benchmark markup $\mu_{w,n,t}$.

Equation (50): Foreign price-level inflation

$$\pi_t^\star = p_t^\star - p_{t-1}^\star.$$

Foreign inflation is the change in the foreign price level.

Equation (51): Import price-level inflation

$$\pi_{f,t} = p_{f,t} - p_{f,t-1}.$$

Import inflation is the change in the import price level.

Equation (52): Real exchange-rate change

$$d\_q_t = q_t - q_{t-1}.$$

The change in the real exchange rate is the first difference of the real exchange-rate level.

Simulation and OSR Blocks

Active shock

The active `shocks` block sets:

$$\text{stderr}(u_{fdi}) = 0.1.$$

All other shock standard errors are commented out. Therefore,

```
stoch_simul(irf=20) y pi d_exc cr rl
```

produces 20-period impulse responses of output, inflation, nominal depreciation, credit, and lending rates to an FDI shock only, unless additional shocks are uncommented.

Optimal simple rule objective

The policy loss is based on the variances of:

$$y_t, \quad \pi_t, \quad d\_exc_t, \quad cr_t, \quad rl_t.$$

With the weights in the file,

$$w_y = w_\pi = w_{dexc} = w_{cr} = w_{rl} = 0.5.$$

The reported loss is:

$$loss\_opt = 10^5 \left[w_y Var(y) + w_\pi Var(\pi) + w_{dexc} Var(d\_exc) + w_{cr} Var(cr) + w_{rl} Var(rl) \right],$$

with the individual variances also stored for reporting after scaling by 10000.

Optimized policy parameters

The OSR block optimizes:

$$\chi_\pi, \quad \chi_y, \quad \chi_{dexc}, \quad \varepsilon_2, \quad \varepsilon_3.$$

Thus the optimizer chooses the monetary-policy responses to inflation, output, and exchange-rate depreciation, plus the DSTI responses to wage inflation and household credit.

Implementation notes

- The model uses $theta\_rl$ to define κ_{rl}. If Dynare raises an "unknown symbol" error, declare $theta\_rl$ as a parameter.
- The loss weights $w_y, w_\pi, w_{dexc}, w_{cr}, w_{rl}$ are assigned after the model block. If Dynare requires explicit declarations in your version, declare them as parameters.
- Since $\lambda_{5,hh} > 0$, $dsti_t$ is best interpreted as an allowed DSTI ceiling or credit-easing measure, not as pure policy tightness.